An Exploratory Data Analysis Report on Public Transit Operations: Insights for MetroMove Transit Solutions

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Introduction

- MetroMove Transit Solutions operates across multiple cities
- Manage thousands of daily trips via buses, trains, ferries, and trams
- Focus on delivering efficient, affordable, and data-driven transportation

Problem Statement

- MetroMove has extensive trip data
- Difficulty extracting meaningful insights due to poor data quality data including messy, inconsistent, and incomplete records
- Limited ability to identify inefficiencies and improve operations

Objectives

- Clean the dataset to ensure data quality
- Analyze passenger usage patterns
- Evaluate performance across transport modes
- Perform descriptive statistics to summarize key metrics
- Create visualizations that effectively communicate findings
- Clear documentations of analysis and actionable recommendations

Data Description

- The dataset was provided by 10alytics and comprises MetroMove daily trip records spanning a six weeks period (Jan - Feb 2024)
- Consists of 1,000 rows and 12 columns
- Key variables captured includes trip id, mode of transport, departure station, arrival station, departure time, passenger count, fare amount, trip duration, date, and day of the week
- Tool was Python, leveraging Pandas, Matplotlib, Seaborn, and Missingno for data manipulation, EDA and visualization

Methodology

Data loading and EDA

- The dataset was loaded into Python from an Excel source and initial profiling done to understand the data types, quality, structure and initial statistical summaries

Data cleaning and preprocessing

- Poor data qualities including inconsistencies were standardized, whitespace errors removed, non-informative/useless fields dropped and new columns were created to enhance analysis
- Missing data patterns were assessed and found to exhibit characteristics consistent with MAR mechanism, informing a grouped-based median imputation of missing values

Methodology

Exploratory analysis and visualization

- Visualization techniques were used to uncover patterns in passenger behaviour, fare structure, and trip performance across transport modes and time variable

Statistical validation

- Post-cleaning descriptive statistics was computed to establish accurate values for key operational metrics
- **Actionable Recommendation**
- Actionable Recommendations informed by findings was made to improve passenger experience and optimize operations

Results

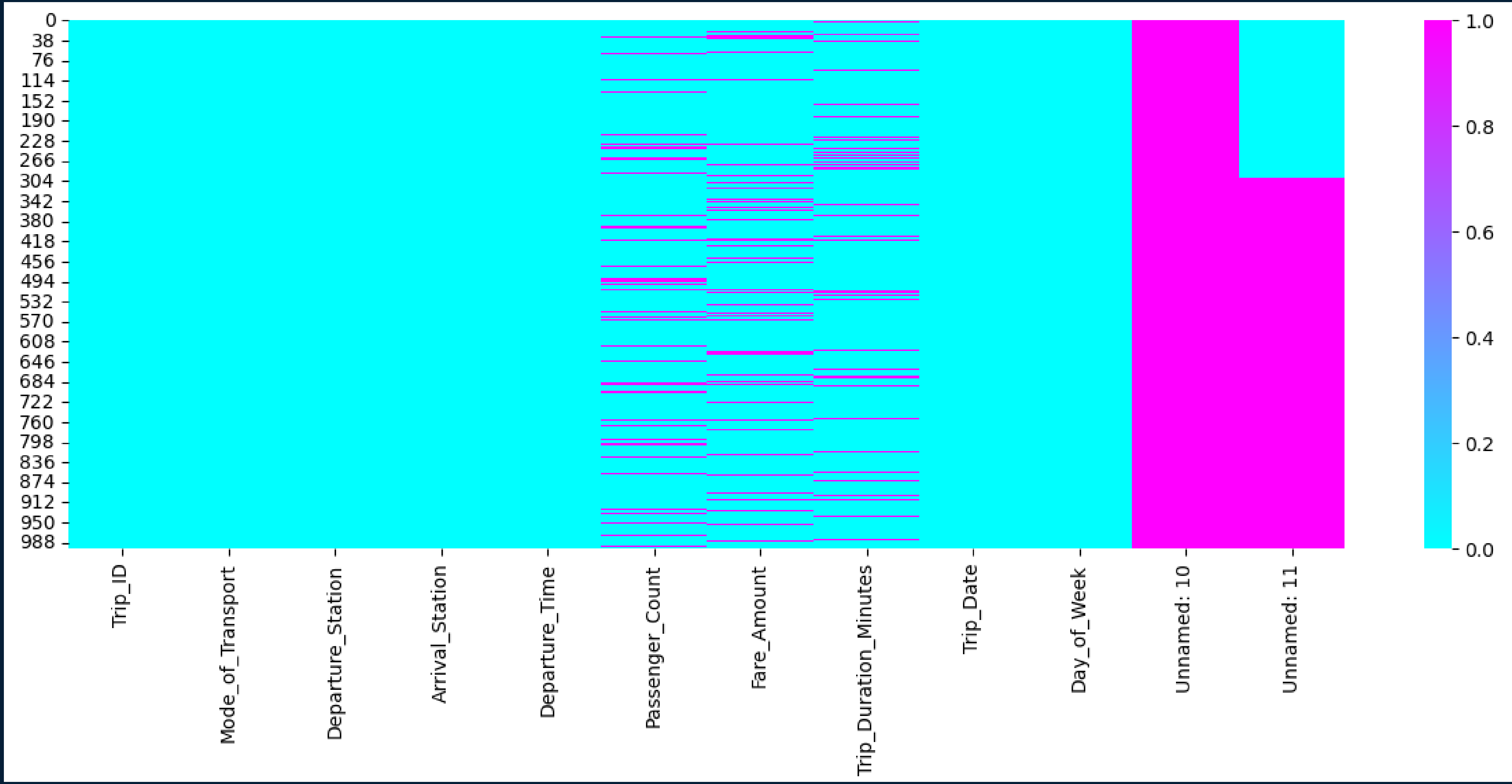

Figure 1a: Heatmap of Dataset Before Data Cleaning

Results

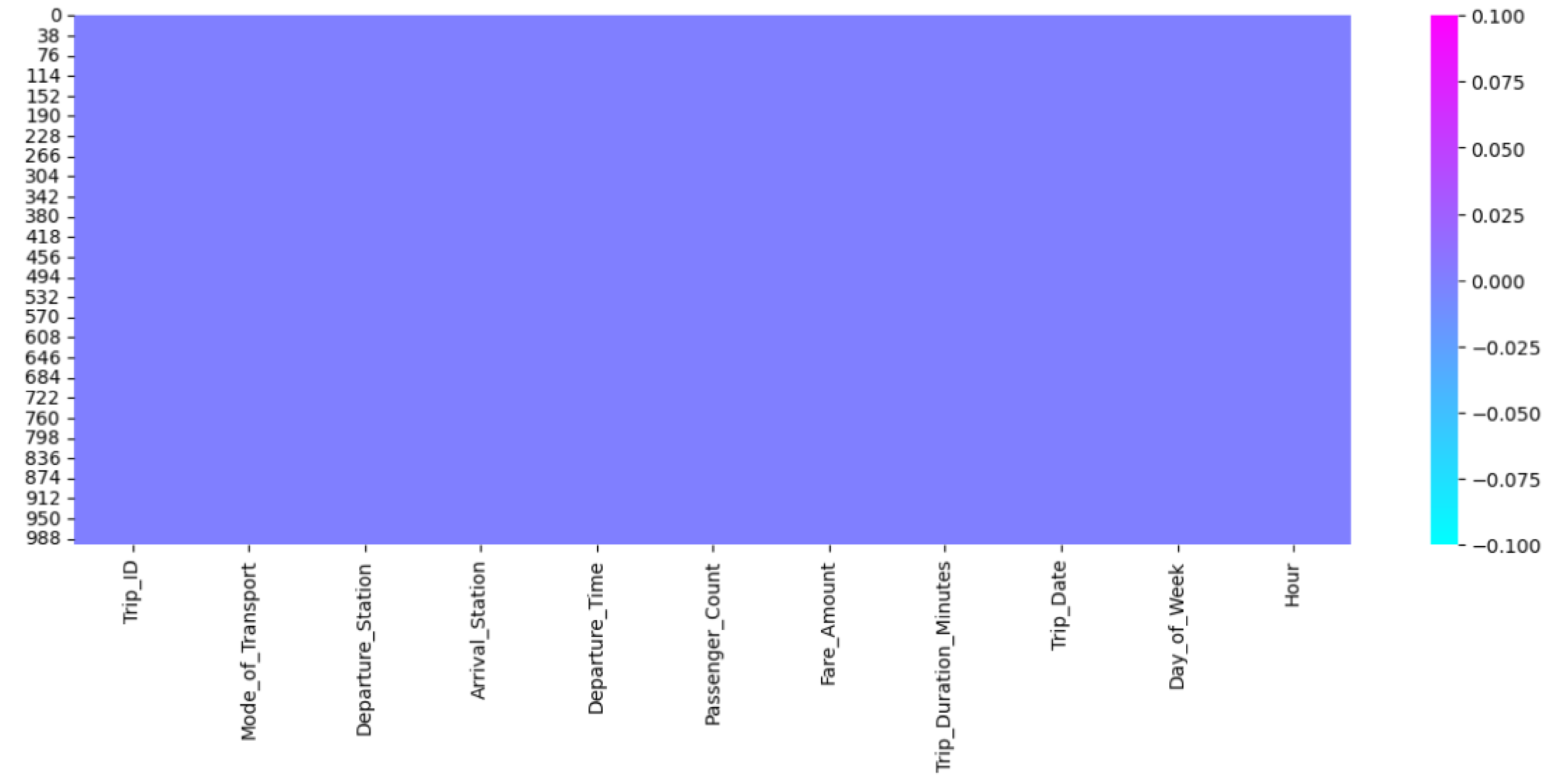

Figure 1b: Heatmap After Data Cleaning

Result

- Key metrics:
  - Total trips 1000
  - Average passengers count 49.5
  - Average fare amount 25.4
  - Average trip duration 94.5

Result

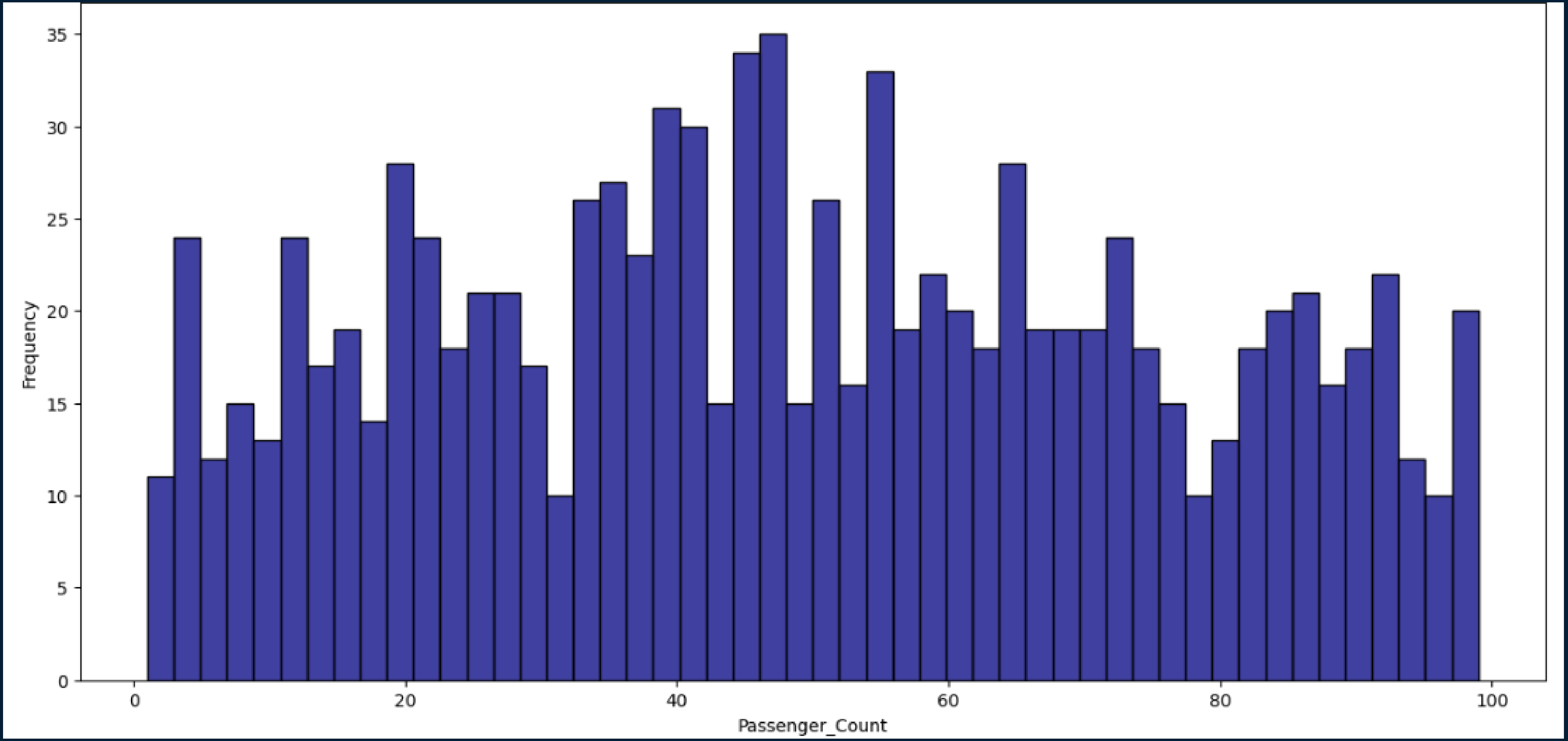

Figure 2: The Total Distribution of Passengers Across All Trips

Result

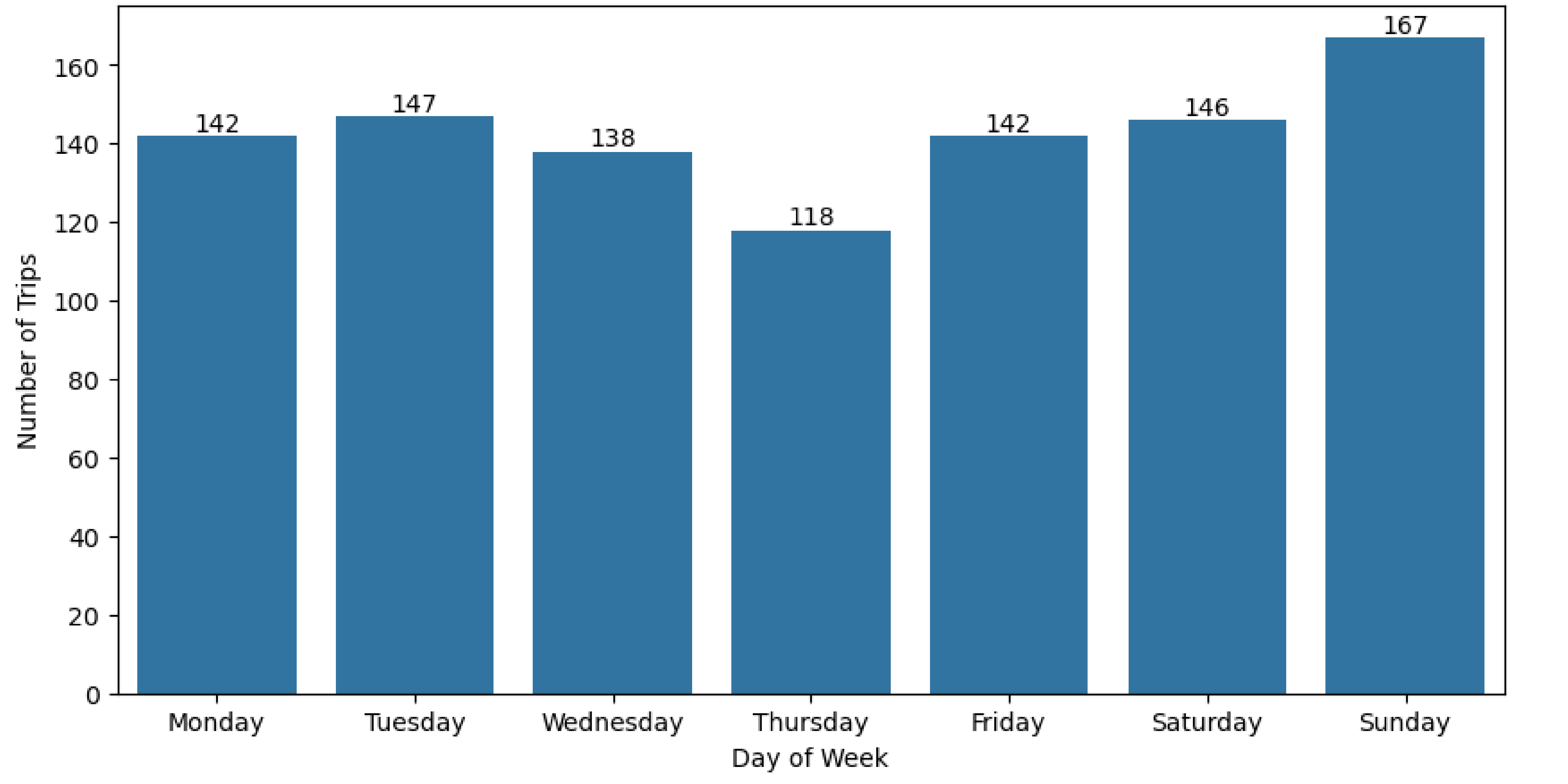

Figure 3: Total Trips by Day of the Week

Results

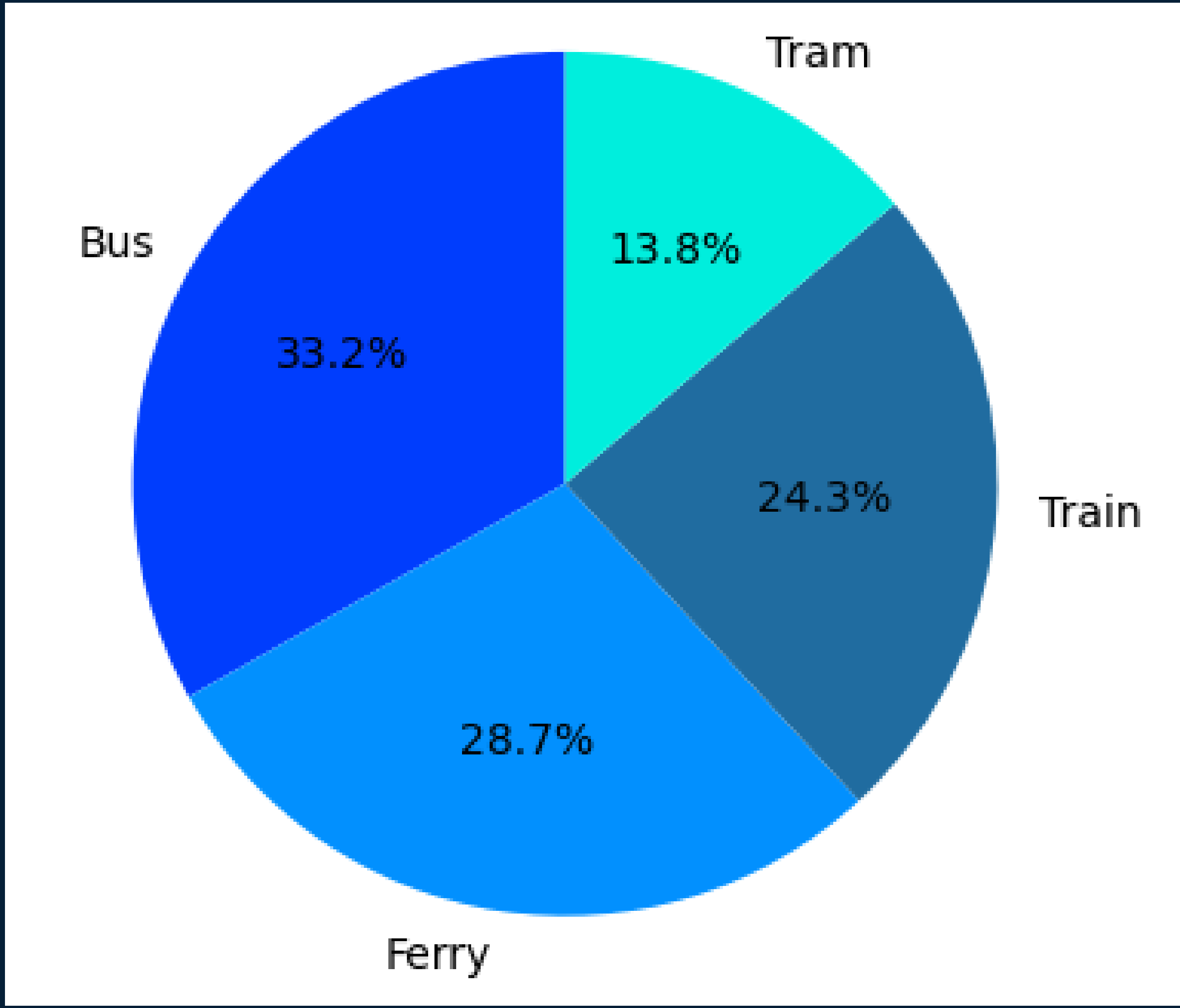

Figure 4: Total Trips by Mode of Transport

Results

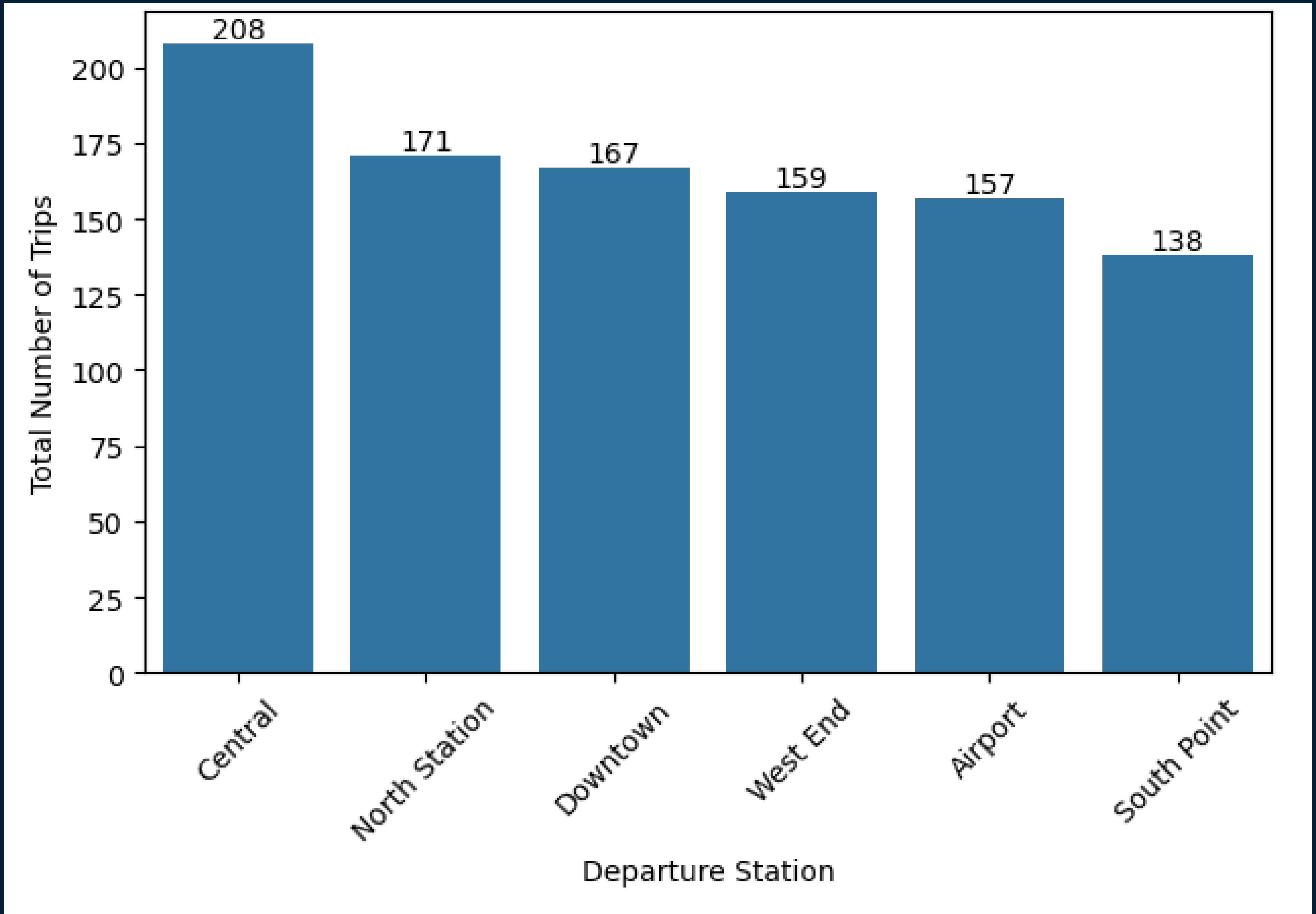

Figure 5: Total Trips by Departure Station

Results

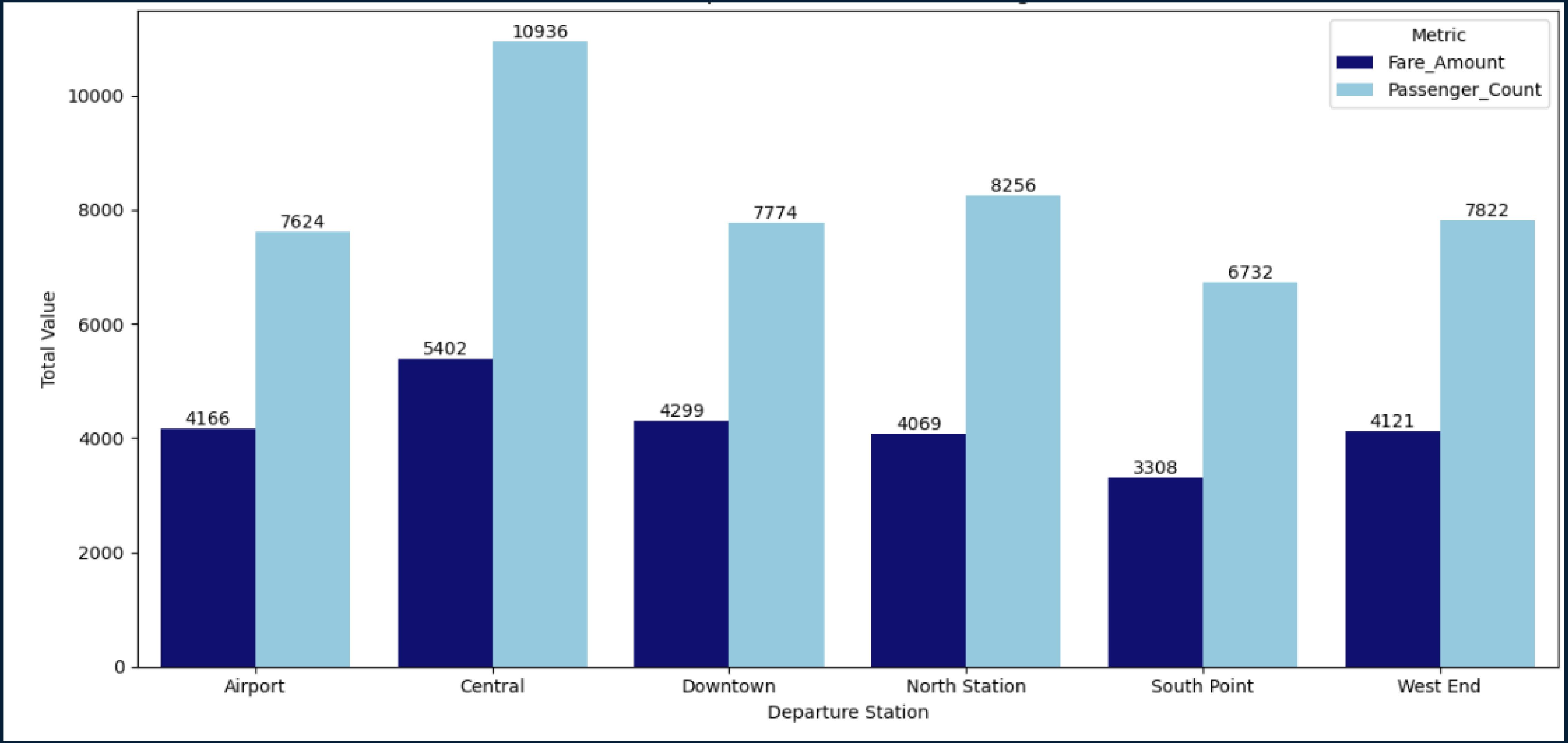

Figure 6: Total Passengers And Total Revenue By Departure Station

Results

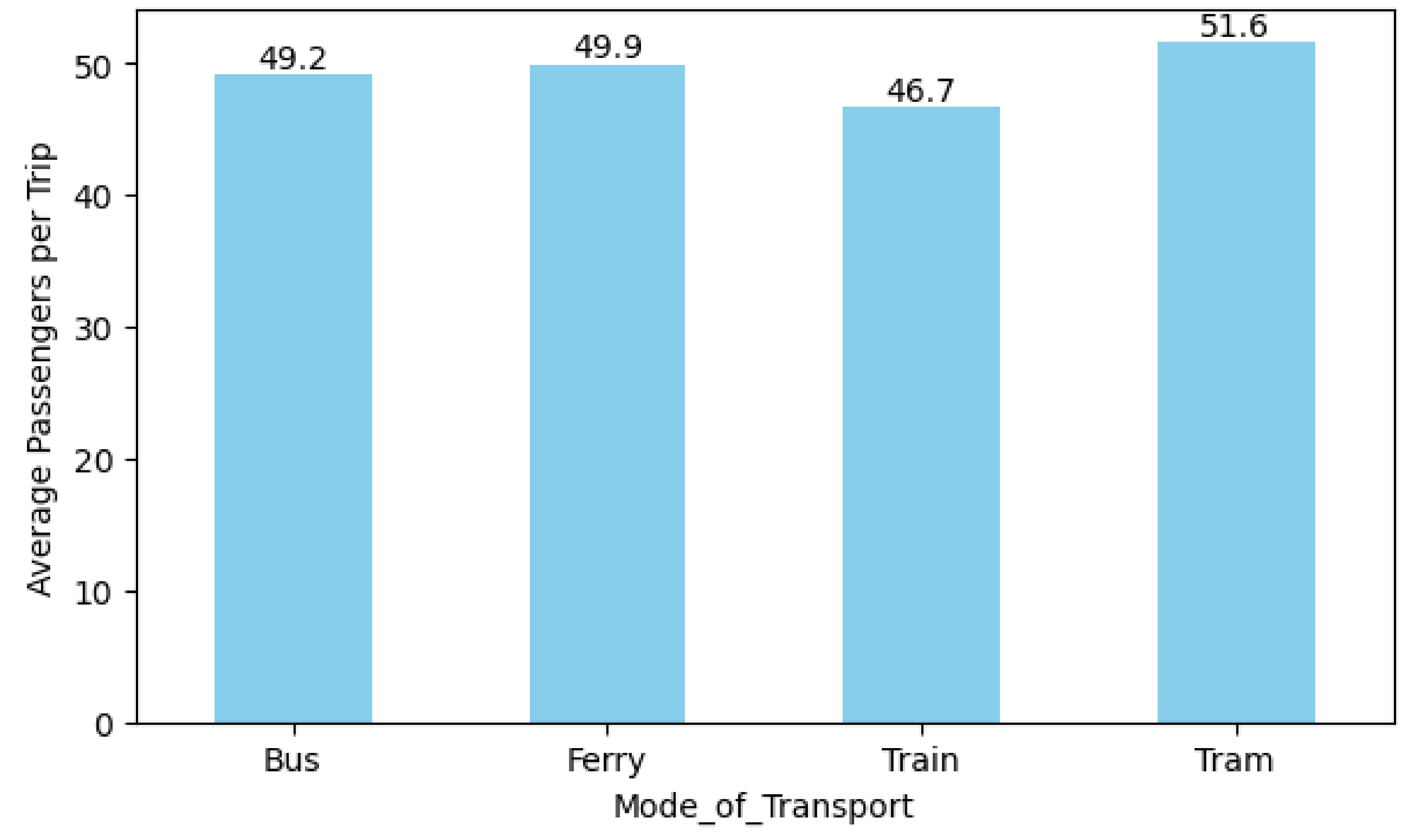

Figure 7: Average Passengers Count Per Trip by Mode of Transport

Results

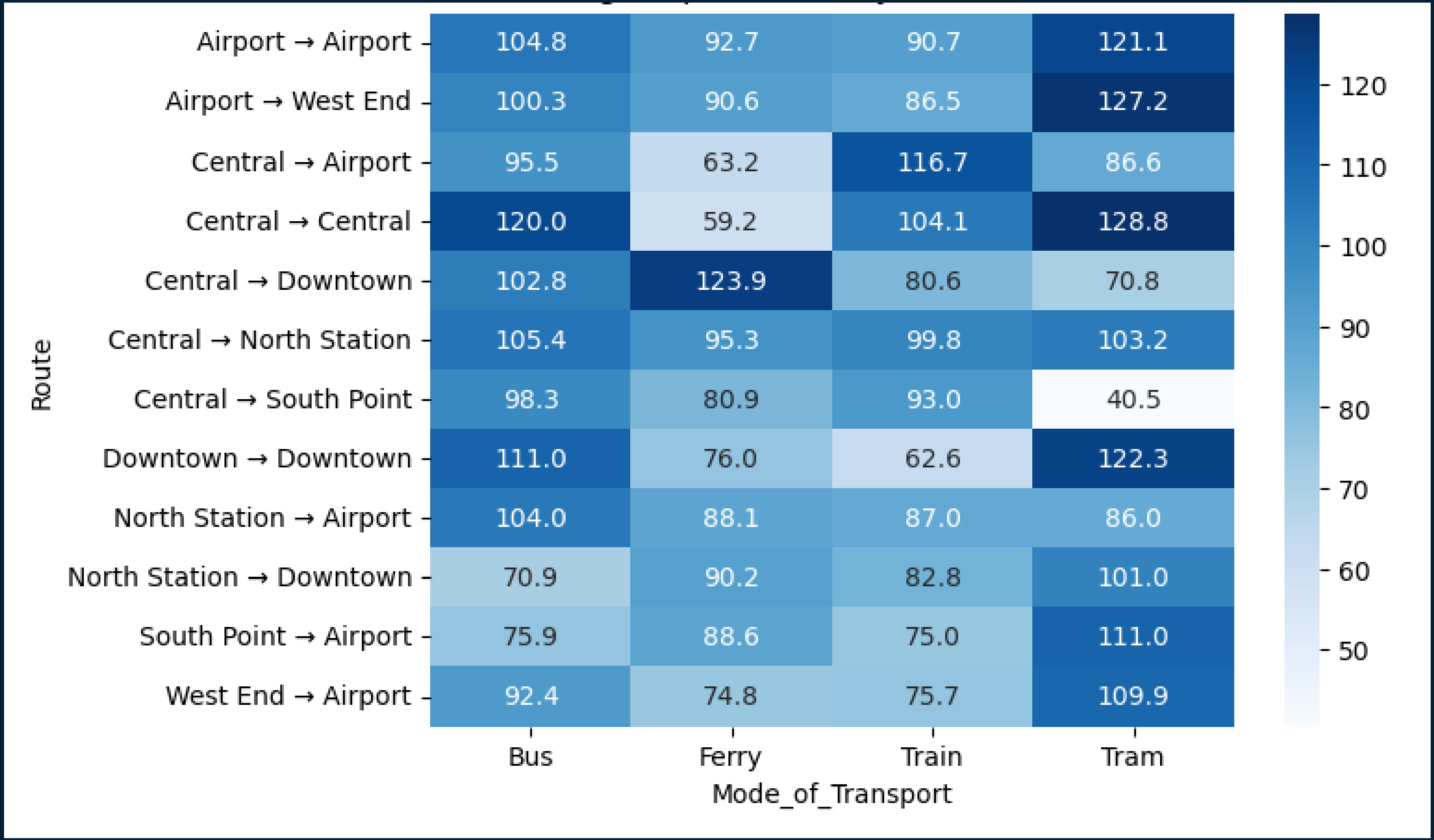

Figure 8: Average Trip Duration by Route and Transport Mode (Performance Comparison)

Results

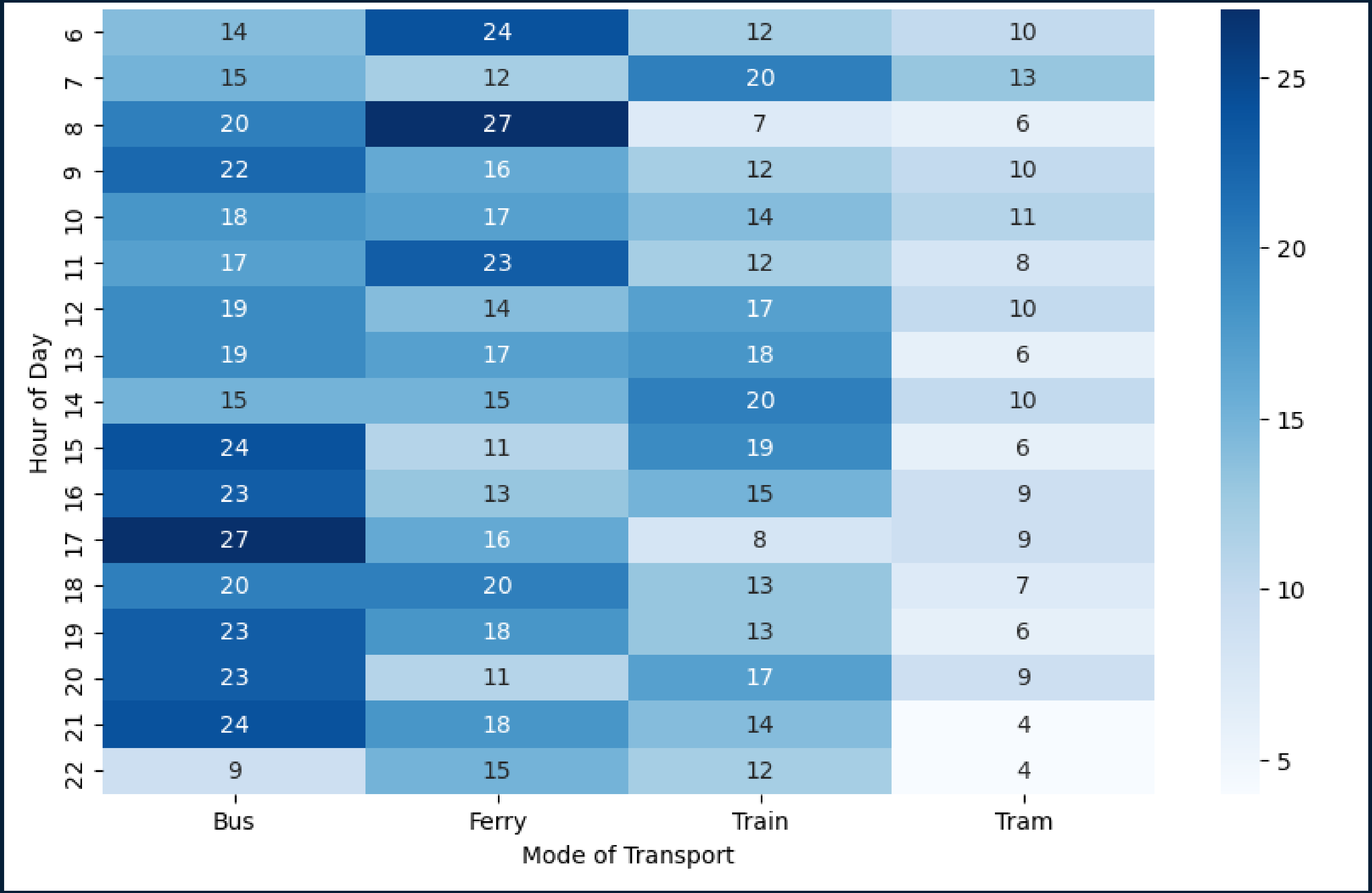

Figure 9: Transport Demand by Mode and Hour of Day

Results

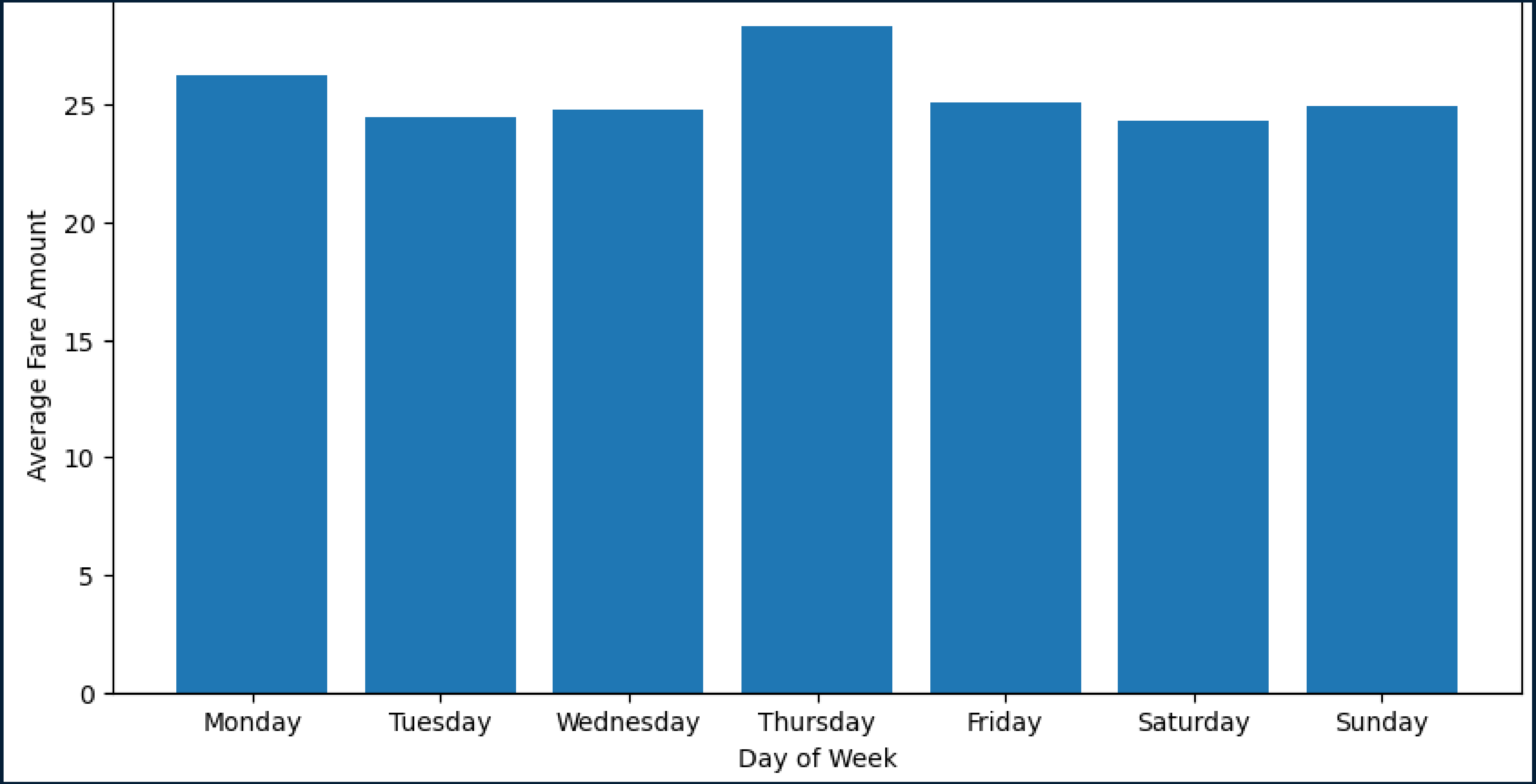

Figure 10: Fare Amount by Day of Week

Results

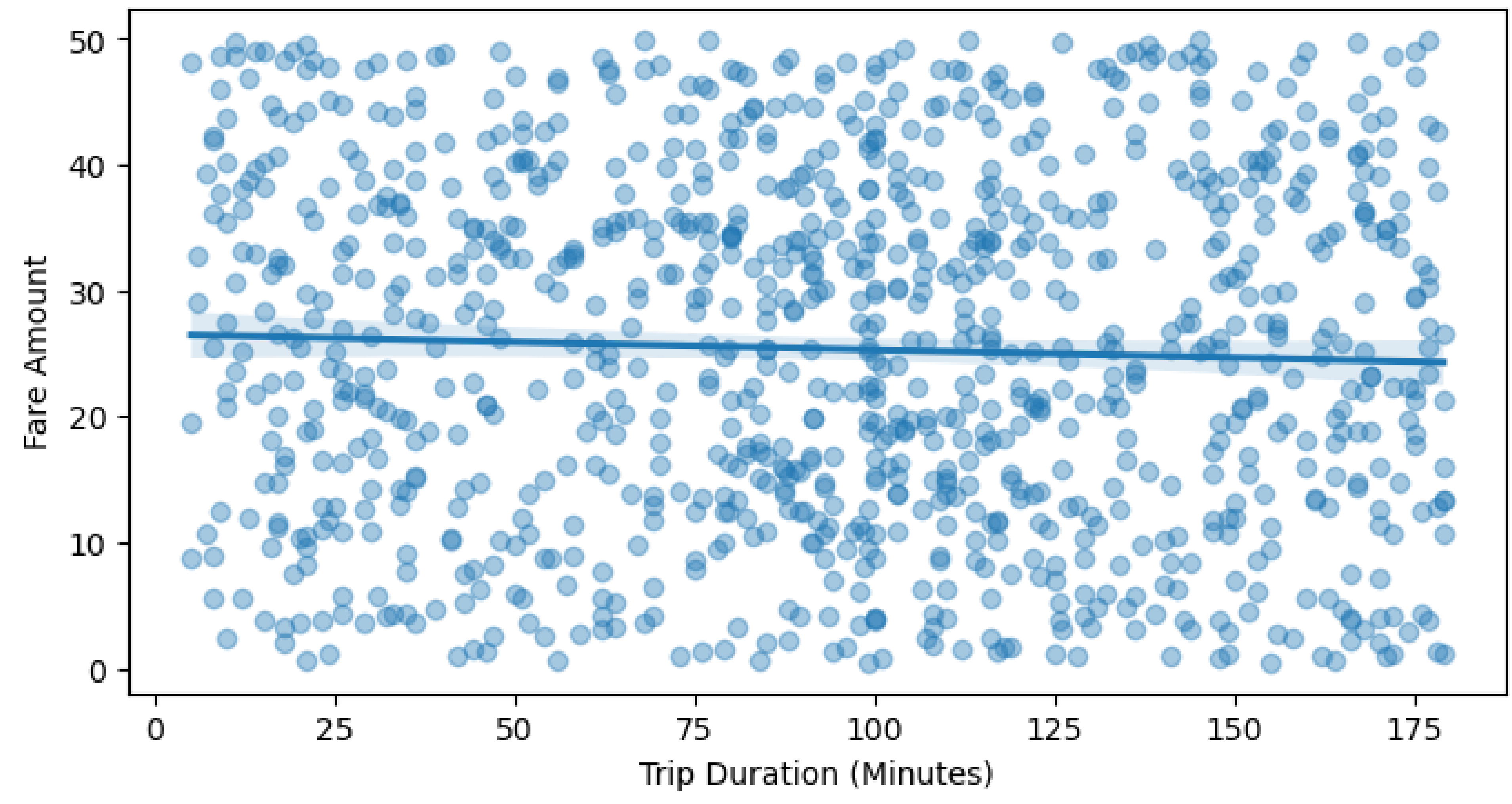

Figure 11: Fare Amount by Trip duration in Minutes

Results

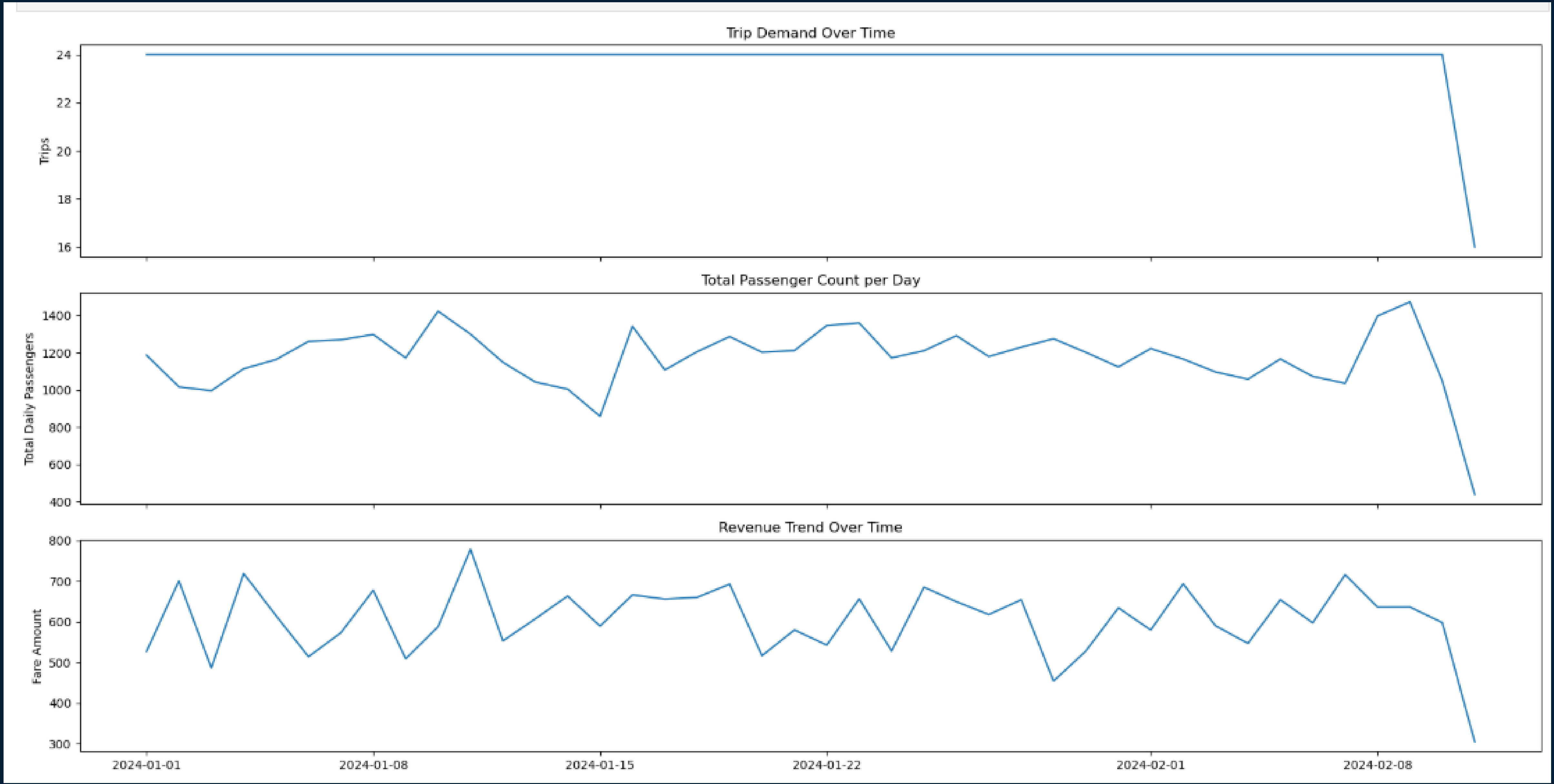

Figure 12: Operational Performance Trends: Trips, Passengers, and Revenue Over Time

Result

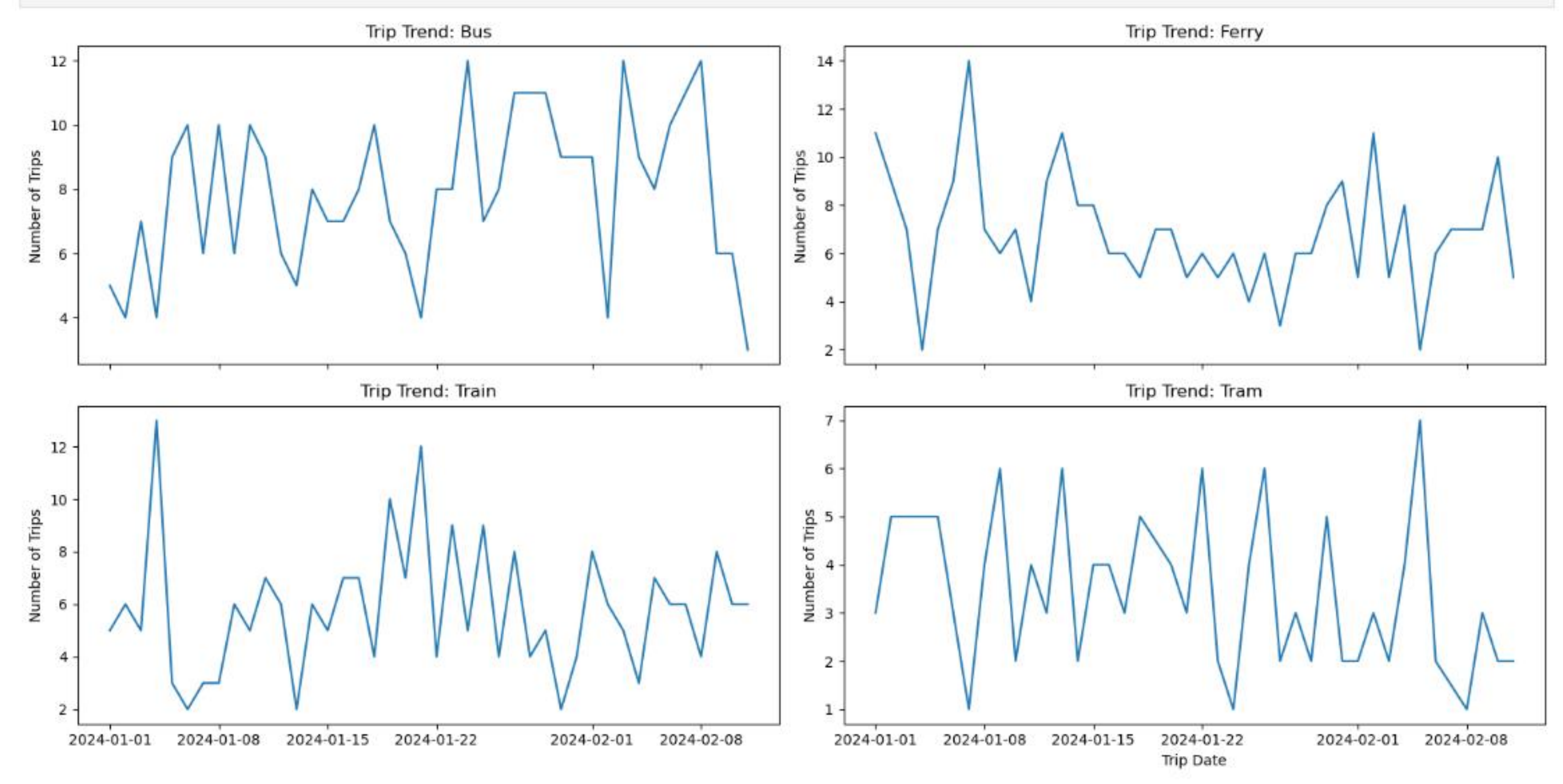

Figure 12: Trends of Passenger Usage Pattern: Mode of Transport

Actionable Recommendations

1. Capacity & Scheduling

- Increase bus capacity during late afternoon and evening (5–9 PM) to meet its peak demand
- Optimize ferry scheduling around the 8 AM morning peak
- Deploy more trains around 7 AM and 2 PM when demand spikes
- Trim tram scheduling and use targeted incentives to address consistent underutilization across all hours
- Use historical trip data to anticipate demand shifts and adjust service levels accordingly

Actionable Recommendations

2. Route Optimization

- Investigate train operations on underperforming routes
- Deploy trains only on routes where they hold a genuine speed advantage
- Introduce express train services on routes where trains are currently slower than buses and trams to restore their expected speed advantage

Actionable Recommendations

3. Pricing & Demand

- Align fares with demand especially on Thursdays with the lowest demand and highest revenue
- Implement targeted interventions at South Point e.g. discounted fares or loyalty rewards to stimulate usage

Limitations

- Short time window (only 6 weeks), which may not capture seasonal demand patterns
- Small dataset (1,000 rows) limits the reliability of trends and generalizations across routes and modes
- No external variables e.g. factors like weather, public holidays, or service disruptions were not captured, which could explain some demand fluctuations
- No passenger demographics; analysis couldn't account for who is travelling (to target incentives more precisely e.g. student discounts, senior passes), limiting equity and targeting insights

Thank you